

3 x 660 MW Supercritical Thermal Project at TPS Koradi



Green Hydrogen Plant at TPS Bhusawal.

of the 1,350 MW Sinnar Thermal Power Plant, the first time a state utility has secured a private thermal project through competitive bidding. The message is clear: this isn't about ideology; it's about securing every megawatt Maharashtra needs.

THE GREEN SURGE

But thermal expansion is merely the foundation for Mahagenco's real ambition. By 2030, the company plans to add over 8 GW of renewable capacity, scaling to 12 GW by 2035. To put that in perspective, it's equivalent to adding a mid-sized European country's entire power capacity, in renewables alone.

The strategy reveals sophisticated thinking about India's energy transition challenges. The "Mukhyamantri Saur Krushi Vahini Yojana 2.0" deploys 1,071 MW of decentralized solar projects within 5-10 km of agriculture-dominated substations, ensuring farmers receive reliable daytime power precisely when they need it for irrigation. It's renewable energy designed around human needs, not abstract targets.

Then there's the water dimension. Mahagenco is developing 3,225 MW of Pumped Storage Plants across sites including Koyna, Ghatghar, Panshet, Varasgaon, and Kodali. These aren't just storage facilities, they're the shock absorbers of the renewable revolution, capturing surplus solar power during peak generation and releasing it when the sun sets and demand spikes. PSPs also provide critical grid services like frequency regulation and black-start capability, making them indispensable for integrating intermittent renewables at scale.

Innovation extends beyond conventional solutions. Floating solar projects totaling 605 MW are planned for sites like Irai Dam and Lower Wardha, overcoming Maharashtra's acute land scarcity. Also, Mahagenco is exploring installation of Battery Energy Storage Systems at various available locations. Mahagenco has signed agreements with NPCIL and MITRA for nuclear development, plus a collaboration with Russia's ROSATOM on thorium-based Small Modular Reactors, aligning with India's indigenous nuclear fuel strategy.



Koyana Hydro Power Station

THE HYDROGEN BET

Perhaps most intriguing is Mahagenco's Green Hydrogen initiative. Following a successful pilot at Bhusawal, the company is planning 50 KTPA (thousand tonnes per annum) projects in the MMRDA region and Uran. This aligns with Maharashtra's Green Hydrogen Policy targeting 500 KTPA by 2030 and positions the state to decarbonize hard-to-abate sectors like refining, fertilizers, and heavy transport, areas where direct electrification remains impractical.

The company has established MAHAGENCO Renewable Energy Ltd (MREL) as a dedicated subsidiary to accelerate this transition, with partnerships including SJVN Limited and NTPC Green Energy Ltd. MREL is separately developing another 2,110 MW of pumped storage capacity and pursuing solar projects with RECPDCL.

ENVIRONMENTAL STEWARDSHIP

Mahagenco's sustainability commitment extends beyond generation mix. The company is implementing Flue Gas Desulphurization across its thermal fleet and retrofitting Electrostatic Precipitators to control air pollution. A Carbon Footprint Study at Koradi Thermal Power Station systematically quantifies emissions as a precursor to reduction strategies.

Water conservation follows the "three R's"—Reduce, Reuse, Recycle. Partnerships with Nagpur Municipal Corporation have established sewage treatment plants supplying 130 MLD to Koradi, 190 MLD split between Koradi and Khaperkheda, and 50 MLD to Chandrapur. Similar facilities are planned at Bhusawal, Parli, and Paras. The goal: "Zero Water Discharge" through maximum recycling. State-of-the-art pipe conveyor systems now transport coal at Chandrapur and are completed for Koradi and Khaperkheda from their respective mines, drastically reducing pollution and coal loss associated with traditional road transport. Biomass co-firing in pulverized coal boilers directly reduces CO2 emissions while conserving coal resources.

THE DIGITAL BACKBONE

None of this works without sophisticated digital infrastructure. Mahagenco's IT Department earned the SAP ACE Award for Public Services for its enterprise resource planning deployment. Real-Time Generation View provides management instantaneous visibility across all generation assets. A Fully Integrated Security Surveillance system protects critical infrastructure.

Most crucial is cybersecurity. Recognizing that power infrastructure represents a prime target for state and non-state actors, Mahagenco has implemented comprehensive controls covering network security, endpoint protection, access control, patch management, and data protection, reinforced by a Security Operation Centre monitoring threats in real-time.

SECURING THE FUEL

Even as it embraces renewables, Mahagenco isn't leaving thermal generation vulnerable to fuel supply disruptions. The Gare Palma-II captive coal mine in Chhattisgarh is projected to reach 23.06 MMT annual production by 2030, eliminating dependence on expensive "Bridge Linkage" coal and securing the base-load fuel supply that underpins grid stability.

THE MAHARASHTRA MODEL

What emerges from Mahagenco's strategy is a coherent theory of energy transition for large, industrialized economies: maintain and modernize thermal capacity as the reliability anchor while aggressively scaling renewables and storage. It's pragmatic environmentalism, recognizing that blackouts serve neither economic growth nor climate goals.

Maharashtra's government plans a phased equity listing of state-owned energy utilities, with the transmission company first, followed by generation and distribution. The move toward capital markets discipline could sharpen efficiency further.

As India pursues Net Zero by 2070 and Maharashtra targets 2047, Mahagenco's roadmap offers a tangible vision: 56% renewable capacity, comprehensive environmental safeguards, and unwavering reliability. It's not merely about generating electricity. It's about generating confidence that India's energy transition can be ambitious and achievable simultaneously—a model that resonates far beyond Maharashtra's borders, to every emerging economy grappling with the same fundamental question: How do you power growth while protecting the planet? In Mahagenco's answer may lie the template for India's and perhaps the world's energy future.



A Memorandum of Understanding signed between MAHAGENCO and Nuclear Power Corporation of India Limited (NPCIL) for developing nuclear power-based electricity generation projects in Maharashtra.



Memorandum of Understanding signed between the Water Resources Department and MAHAGENCO for Koyana Stage-VI Pumped Storage Plant.